

BC Experiment - 10

Aim: Explore the Cryptocurrency Landscape

Theory:

This experiment explores the **cryptocurrency landscape** by combining real-world market data analysis with blockchain technology research. The focus asset for this analysis is **Chainlink (LINK)**, and data was collected from **three reputable sources** as per the lab requirement:

1. **CoinGecko API** – Provided historical price data, market capitalization, and trading volumes in a structured JSON format, enabling direct processing in Python.
2. **CoinMarketCap** – Offered real-time and historical price charts, circulating supply, and market rank for visual cross-verification.
3. **Crypto.com** – Supplied quick-access market statistics, simplified price trends, and cryptocurrency news updates.

Cryptocurrency Landscape

1. **Definition** – Cryptocurrencies are decentralized digital assets built on blockchain networks, enabling secure, transparent, and immutable transactions without central authority.
2. **Market Diversity** – Thousands of cryptocurrencies exist, each serving unique roles:
 - **Store of Value** – Bitcoin, Litecoin.
 - **Utility Tokens** – Chainlink, Ethereum.
 - **Stablecoins** – USDT, USDC.
 - **Governance Tokens** – UNI, AAVE.
3. **Volatility** – Prices are highly sensitive to market sentiment, technological developments, and regulatory updates.
4. **Data Analysis Importance** – Tracking historical trends, market capitalization, and trading volume is essential for understanding asset performance.
5. **Advancements** – Growing use of **DeFi platforms**, **NFT ecosystems**, **layer-2 scaling**, and **cross-chain protocols**.

Data Collection Process

The program used the **CoinGecko API** to automate historical data fetching for Chainlink. The retrieved data was processed with **Pandas** to compute:

- **Daily Returns** – Percentage change between consecutive days to measure volatility.
- **Moving Averages (MA7, MA30)** – Short-term and long-term trend indicators.
- **Cumulative Returns** – Growth factor over time.
- **Rolling Volatility** – Short-term risk measurement based on daily return fluctuations.

Visualization & Dashboard

Using **Matplotlib**, a six-panel dashboard was created to display:

1. **Price Trend** – Daily price movement over the selected period.
2. **Price vs Volume** – Correlation between trading activity and price changes.
3. **Daily Returns Distribution** – Statistical spread of daily gains/losses.

4. **Moving Averages** – Technical analysis trend lines for price movement.
5. **Cumulative Returns** – Overall growth performance.
6. **Rolling Volatility** – Market risk trends over time.

Technological Analysis

Beyond market data, the study covered **advancements in blockchain technology**, including:

- **Consensus Mechanisms** – PoW (Proof of Work), PoS (Proof of Stake), and hybrid systems.
- **Scalability Solutions** – Layer-2 protocols, sharding, and sidechains to increase throughput.
- **Interoperability Protocols** – Chainlink’s CCIP, Polkadot, and Cosmos enabling cross-chain data exchange.
- **Privacy Features** – Zero-knowledge proofs (zk-SNARKs) and ring signatures to ensure transaction confidentiality.

Chainlink’s **roadmap** highlights staking implementation, advanced oracle features, and enhanced cross-chain capabilities, aligning with industry trends toward **decentralization, interoperability, and security**.

Market Trends & Adoption

- **Growing Merchant Acceptance** – More businesses accepting cryptocurrency payments.
- **Wallet Growth** – Increasing downloads and installations of crypto wallets.
- **Network Activity** – Higher transaction volumes and active addresses, signaling strong adoption.
- **Integration with Traditional Finance** – Partnerships between crypto platforms and fintech/payment providers.
- **Regulatory Developments** – Countries exploring frameworks for safe cryptocurrency adoption.

Implementation

coingecko.com/en/all-cryptocurrencies

Now LIVE: Spot CEX Report 2026

Coins: 17,727 Exchanges: 1,457 Market Cap: \$2.52T 0.3% 24h Vol: \$80.418B Dominance: BTC 57.1% ETH 10.5% Gas: 0.195 GWEI

Go Ad-free

coingecko

Cryptocurrencies Exchanges NFT Learn Products API

Candy Portfolio Search

Market Cap

24h Volume

24h Change

Price

Category

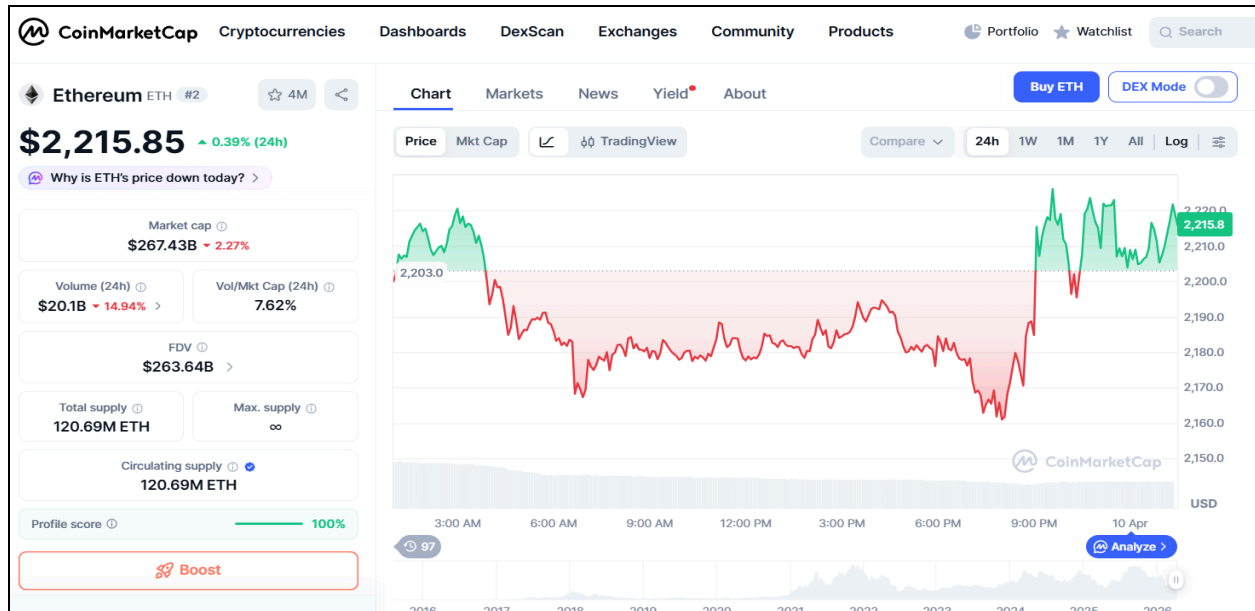
Exchange

Platform

Search

Reset

#	Coin	Price	1h	24h	7d	30d	24h Volume	Circulating Supply	Total Supply	Market Cap
1	Bitcoin BTC	\$71,331.17	▲ 1.6%	▲ 0.9%	▲ 7.4%	▲ 2.8%	\$32,137,027,517	20,013,568	20.01M	\$1,428,416,484,857
2	Ethereum ETH	\$2,183.85	▲ 1.7%	▼ 0.5%	▲ 6.7%	▲ 8.3%	\$14,281,980,651	120,691,116	120.69M	\$263,733,056,490
3	Tether USDT	\$1	▲ 0.0%	▼ 0.0%	▲ 0.0%	▼ 0.0%	\$52,893,681,572	184,112,071,794	189.58B	\$184,108,089,724
4	XRP XRP	\$1.35	▲ 1.6%	▼ 0.7%	▲ 3.4%	▼ 3.6%	\$2,275,850,226	61,405,531,717	99.99B	\$82,155,932,063
5	BNB BNB	\$601.27	▲ 0.7%	▼ 0.2%	▲ 4.8%	▼ 6.0%	\$860,423,604	136,356,689	136.36M	\$82,048,486,225
6	USDC USDC	\$1	▼ 0.1%	▼ 0.1%	▼ 0.1%	▼ 0.1%	\$11,525,970,626	78,289,230,299	78.31B	\$78,288,705,069



Utilizing Api for analysis and visualization : Code:

```
import requests
```

```
import matplotlib.pyplot as plt
```

```
import pandas as pd
```

```
# ----- Fetch Data -----
```

```
def get_historical_data(coin_id, days='180', currency='usd'):
```

```
    url = f'https://api.coingecko.com/api/v3/coins/{coin_id}/market_chart'
```

```
    params = {'vs_currency': currency, 'days': days}
```

```
    response = requests.get(url, params=params)
```

```
    return response.json()
```

```
# ----- Dashboard Function -----
```

```
def ethereum_dashboard(days='180'):
```

```
    coin_id = 'ethereum'
```

```
    data = get_historical_data(coin_id, days)
```

```
# Create DataFrame
```

```

prices_df = pd.DataFrame(data['prices'], columns=['timestamp', 'price'])
volumes_df = pd.DataFrame(data['total_volumes'], columns=['timestamp', 'volume'])

prices_df['timestamp'] = pd.to_datetime(prices_df['timestamp'], unit='ms')
volumes_df['timestamp'] = pd.to_datetime(volumes_df['timestamp'], unit='ms')

df = pd.merge(prices_df, volumes_df, on='timestamp')
df.set_index('timestamp', inplace=True)

# Calculate extra metrics
df['returns'] = df['price'].pct_change()
df['MA7'] = df['price'].rolling(window=7).mean()
df['MA30'] = df['price'].rolling(window=30).mean()
df['cumulative'] = (1 + df['returns']).cumprod()
df['volatility'] = df['returns'].rolling(window=7).std() * 100

# ----- Plot Dashboard -----
fig, axes = plt.subplots(3, 2, figsize=(16, 12))
fig.suptitle(f'{coin_id.capitalize()} Dashboard (Last {days} Days)', fontsize=16, fontweight='bold')

# 1. Price Trend
axes[0, 0].plot(df.index, df['price'], color='blue')
axes[0, 0].set_title('Price Trend')
axes[0, 0].set_ylabel('Price (USD)')
axes[0, 0].set_xlabel('Year-Month') # Add xlabel
axes[0, 0].grid(True)

# 2. Price vs Volume
axes[0, 1].plot(df.index, df['price'], color='blue', label='Price')
axes[0, 1].bar(df.index, df['volume'], color='orange', alpha=0.3, label='Volume')

```

```
axes[0, 1].set_title('Price vs Volume')
axes[0, 1].set_xlabel('Year-Month') # Add xlabel
axes[0, 1].set_ylabel('Price (USD) / Volume') # Add ylabel
axes[0, 1].legend()
axes[0, 1].grid(True)
```

3. Daily Returns Distribution

```
axes[1, 0].hist(df['returns'].dropna(), bins=50, alpha=0.7, color='purple')
axes[1, 0].set_title('Daily Returns Distribution')
axes[1, 0].set_xlabel('Daily Return')
axes[1, 0].set_ylabel('Frequency')
axes[1, 0].grid(True)
```

4. Moving Averages

```
axes[1, 1].plot(df.index, df['price'], color='lightgray', label='Price')
axes[1, 1].plot(df.index, df['MA7'], color='blue', label='7-day MA')
axes[1, 1].plot(df.index, df['MA30'], color='red', label='30-day MA')
axes[1, 1].set_title('Moving Averages')
axes[1, 1].set_xlabel('Year-Month')
axes[1, 1].set_ylabel('Price (USD)')
axes[1, 1].legend()
axes[1, 1].grid(True)
```

5. Cumulative Returns

```
axes[2, 0].plot(df.index, df['cumulative'], color='green')
axes[2, 0].set_title('Cumulative Returns')
axes[2, 0].set_ylabel('Growth Factor')
axes[2, 0].set_xlabel('Year-Month')
axes[2, 0].grid(True)
```

6. Volatility

```

axes[2, 1].plot(df.index, df['volatility'], color='orange')

axes[2, 1].set_title('Rolling Volatility (7-Day)')

axes[2, 1].set_ylabel('Volatility (%)')

axes[2, 1].set_xlabel('Year-Month')

axes[2, 1].grid(True)

plt.tight_layout(rect=[0, 0, 1, 0.96])

plt.subplots_adjust(hspace=0.5, wspace=0.3) # Adjust vertical and horizontal spacing

plt.show()

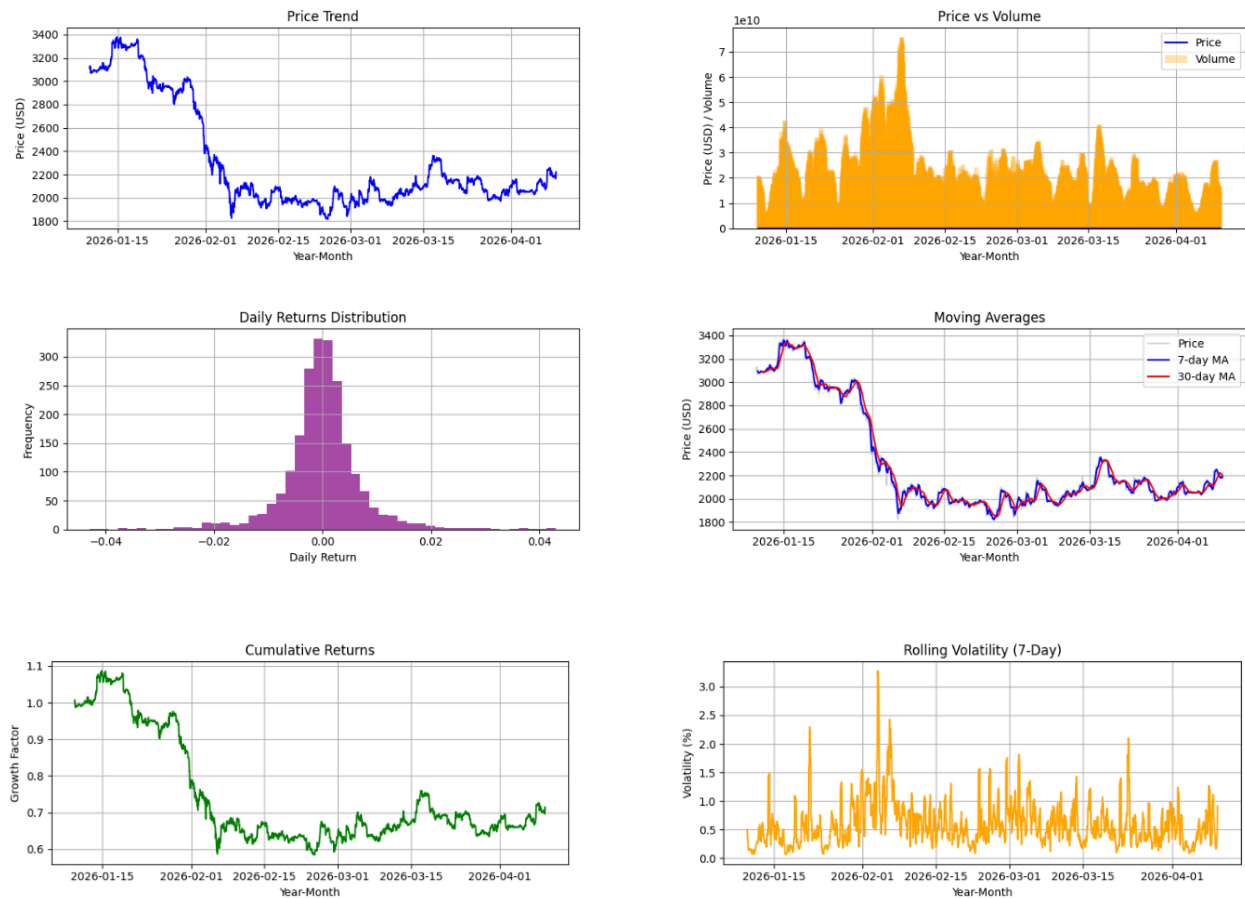
# ----- Run Dashboard -----

ethereum_dashboard('90')

```

Output:

Ethereum Dashboard (Last 90 Days)- Ansh



Conclusion:

By combining automated market data processing, visual trend analysis, blockchain technology evaluation, and adoption metrics, this experiment presents a holistic view of the cryptocurrency ecosystem. It demonstrates how blockchain's technical foundations connect with real-world market behavior, providing insights into both current trends and future advancements in the digital asset industry.